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5CM506 Data driven systems

SQL (Structure Query Language)

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Aims of the Session

This session aims to help you:

1. Understand the basic principles of SQL

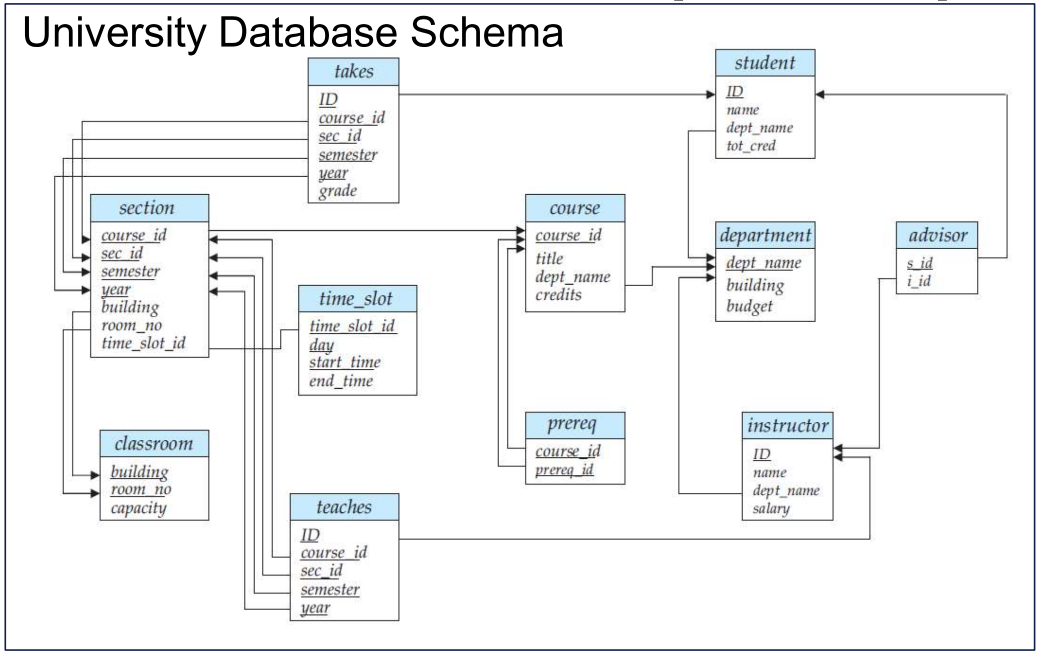
2. Describe the different data types in SQL

3. Apply the theoretical concepts to complete SQL queries and schema definition queries

Overview

- **Recap**
- Advanced Queries
- Data Definition
- Schema Definition

The *insert* Command (recap)



instructor
ID
name
dept_name
salary

ACVITITY: Complete the following queries

We want to add now the following records:

- Sara, with id 787091, dept of Computing, 64000£
- Leonardo, with id 787092, dept of Computing, 55000£

The *insert* Command (recap)

ACVITITY: Complete the following queries

We want to add now the following records:

- Sara, with id 787091, dept of Computing, 64000£
- Leonardo, with id 787092, dept of Computing, 55000£

instructor
<u>ID</u>
name
<u>*dept_name*</u>
salary

insert into *instructor*
  values *(787091, ‘Sara’, ‘Computing’, 64000);*

insert into *instructor*
  values *(787092, ‘Leonardo’, ‘Computing’, 55000);*

	ID	name	dept_name	salary
1	787091	Sara	Computing	64000
2	787092	Leonardo	Computing	55000

The *select* Query (recap)

ACVITITY: Complete the following queries

We want to find the following:

- The department name of all instructors
- The salary of all instructors

instructor
ID
name
dept_name
salary

The *select* Query (recap)

ACVITITY: Complete the following queries

We want to find the following:

- The department name of all instructors
- The salary of all instructors

instructor
ID
name
dept_name
salary

select *dept_name*
from *instructor;*

	dept_name
1	Computing
2	Computing

select *salary*
from *instructor;*

	salary
1	64000
2	55000

Overview

- Recap
- **Advanced Queries**
- Data Definition
- Schema Definition

(Not So) Advanced Queries

ACVITITY: Consider the following queries

- Find the department names of all instructors
- Find the department names of all instructors without repetitions
- Find the id and salary of all instructors in the case of a 10% salary increase

instructor
<u>ID</u>
name
<u>dept_name</u>
salary

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(Not So) Advanced Queries

ACVITITY: Consider the following queries

- Find the department names of all instructors
- Find the department names of all instructors without repetitions
- Find the id and salary of all instructors in the case of a 10% salary increase

instructor
ID
name
dept_name
salary

select *dept_name*
from *instructor;*

	dept_name
1	Computing
2	Computing

(Not So) Advanced Queries

ACVITITY: Consider the following queries

- Find the department names of all instructors
- Find the department names of all instructors without repetitions
- Find the id and salary of all instructors in the case of a 10% salary increase

instructor
ID
name
dept_name
salary

select *dept_name*
from *instructor;*

	dept_name
1	Computing
2	Computing

select distinct *dept_name*
from *instructor;*

	dept_name
1	Computing

(Not So) Advanced Queries

ACVITITY: Consider the following queries

- Find the department names of all instructors
- Find the department names of all instructors without repetitions
- Find the id and salary of all instructors in the case of a 10% salary increase

instructor
ID
name
dept_name
salary

***select** dept_name*
***from** instructor;*

	dept_name
1	Computing
2	Computing

***select distinct** dept_name*
***from** instructor;*

	dept_name
1	Computing

***select** ID, salary*1.1*
***from** instructor;*

	ID	(No column name)
1	787091	70400.0
2	787092	60500.0

(Not So) Advanced Queries

Consider the following query:

- Find the names of all instructors in the Computing department whose salary is greater than 60000£

select *name*
from *instructor*
where *dept_name='Computing'* ***and*** *salary>60000;*

	name
1	Sara

instructor
<u>ID</u>
name
<u>*dept_name*</u>
salary

(Not So) Advanced Queries

Consider the following query:

- Find the names of all instructors in the Computing department whose salary is greater than 60000£

select *name*
from *instructor*
where *dept_name='Computing'* ***and*** *salary>60000;*

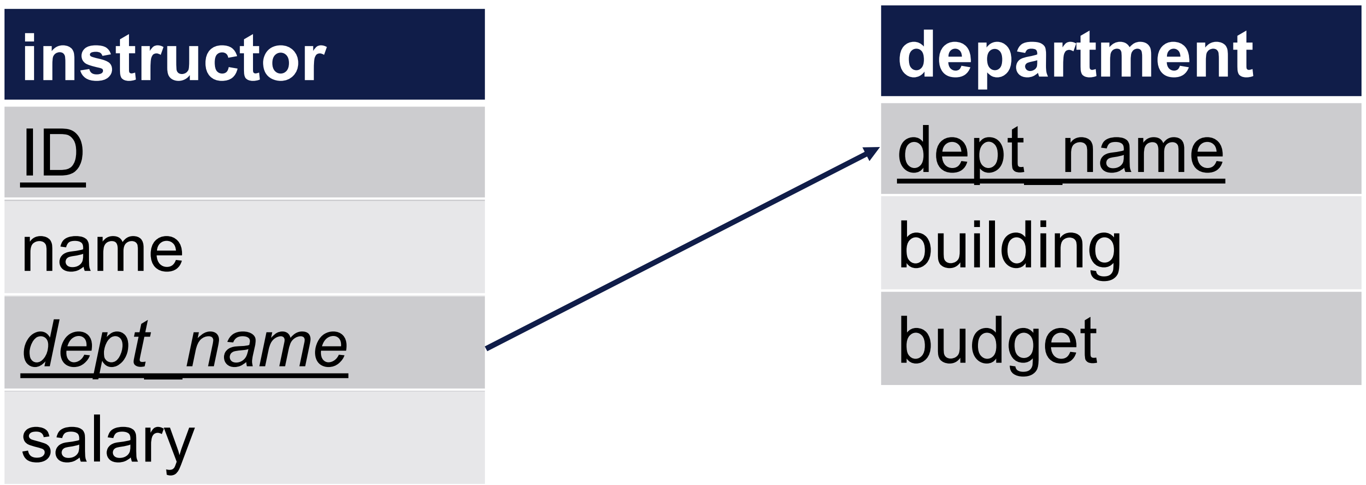
instructor
ID
name
dept_name
salary

SQL allows the use of logical connectives in the ***where*** clause:
and, **or**, **not**
As well as comparison operators:
<, <=, >, >=, =, <>

Advanced Queries

ACVITITY: Consider the following query

- Retrieve the names of all instructors, along with their department names and department building name



	ID	name	dept_name	salary
1	787091	Sara	Computing	64000
2	787092	Leonardo	Computing	55000
3	787093	John	NULL	500

	dept_name	building	budget
1	Computing	Markeaton	1000000
2	Mechanical	Markeaton	500000

Advanced Queries

ACVITITY: Consider the following query

- Retrieve the names of all instructors, along with their department names and department building name

instructor
ID
name
dept_name
salary

department
dept_name
building
budget

select *name, instructor.dept_name, building*
from *instructor, department*

	name	dept_name	building
1	Sara	Computing	Markeaton
2	Leonardo	Computing	Markeaton
3	John	NULL	Markeaton
4	Sara	Computing	Markeaton
5	Leonardo	Computing	Markeaton
6	John	NULL	Markeaton

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Advanced Queries

ACVITITY: Consider the following query

- Retrieve the names of all instructors, along with their department names and department building name

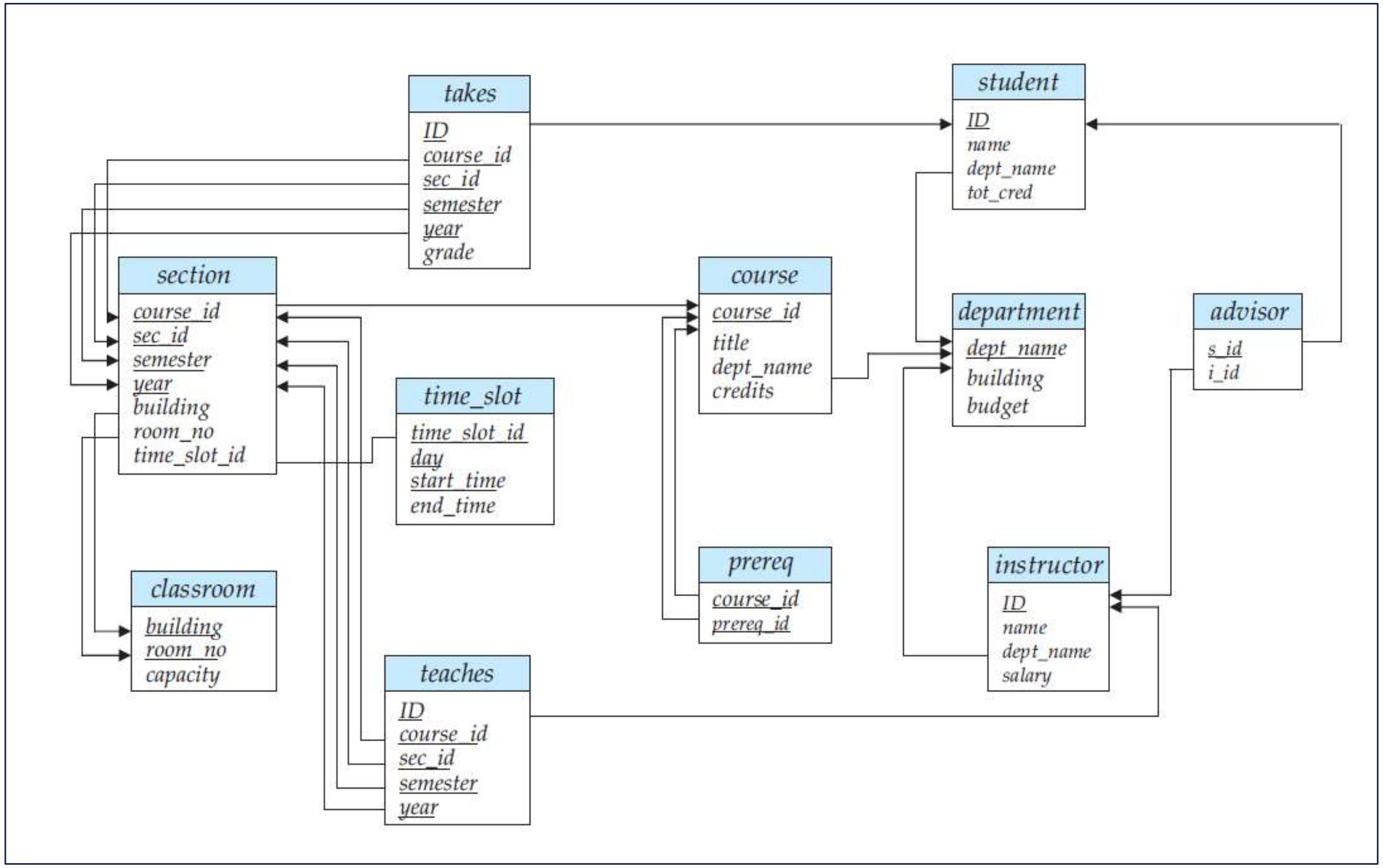
instructor
ID
name
dept_name
salary

department
dept_name
building
budget

```
select name, instructor.dept_name, building
from instructor, department
where instructor.dept_name=department.dept_name;
```

	name	dept_name	building
1	Sara	Computing	Markeaton
2	Leonardo	Computing	Markeaton

Example: University Database



Advanced Queries

ACVITITY: Consider the following query

- For all the instructors in the university that are an advisor for some students, find their names and the student ids that they are an advisor for

student
<u>ID</u>
name
<u>dept_name</u>
tot_cre

advisor
<u>s_id</u>
<u>i_id</u>

instructor
<u>ID</u>
name
<u>dept_name</u>
salary

	s_id	i_id
1	1000	787091
2	1002	787092

	ID	name	dept_name	tot_cred
1	1000	Tom	Computing	200
2	1001	Kate	Mechanical	200
3	1002	Ali	Computing	200

	ID	name	dept_name	salary
1	787091	Sara	Computing	64000
2	787092	Leonardo	Computing	55000
3	787093	John	NULL	500

Advanced Queries

ACVITITY: Consider the following query

- For all the instructors in the university that are an advisor for some students, find their names and the student ids that they are an advisor for

advisor
s_id
i_id

instructor
ID
name
dept_name
salary

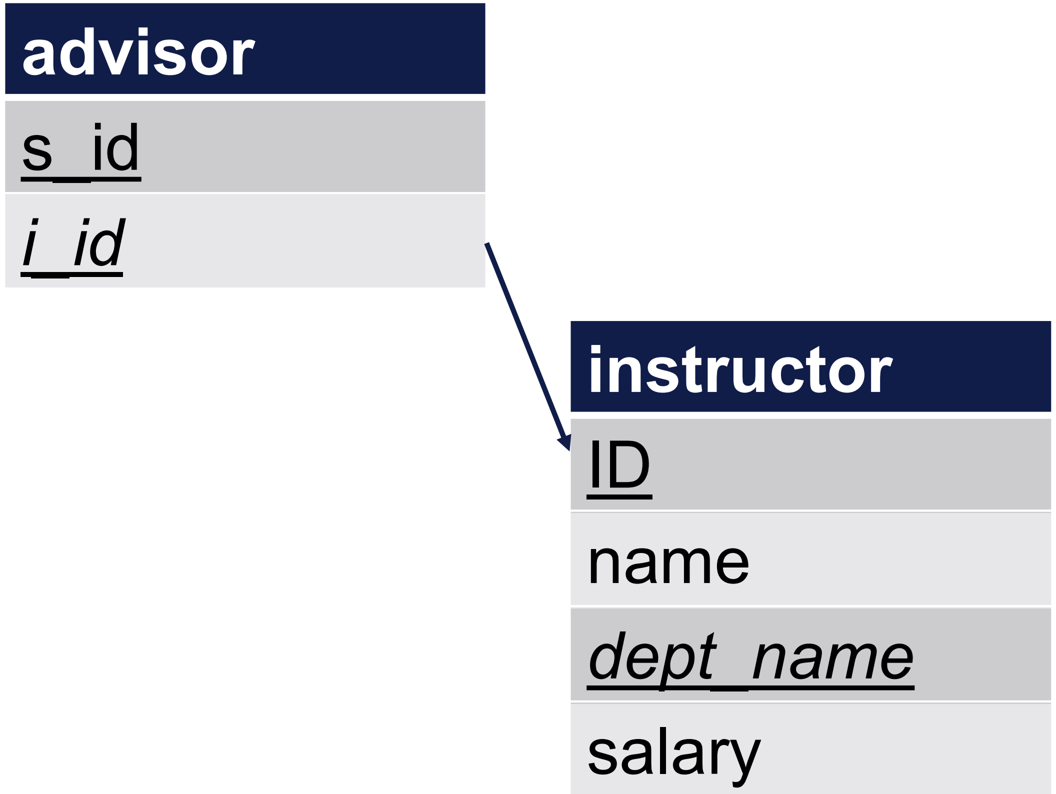
select *name, s_id*
from *instructor, advisor*
where *instructor.ID=advisor.i_id;*

	name	s_id
1	Sara	1000
2	Leonardo	1002

Additional Operations: as

Let us consider the previous query:

- For all the instructors in the university that are an advisor for some students, find their names and the student ids that they are an advisor for



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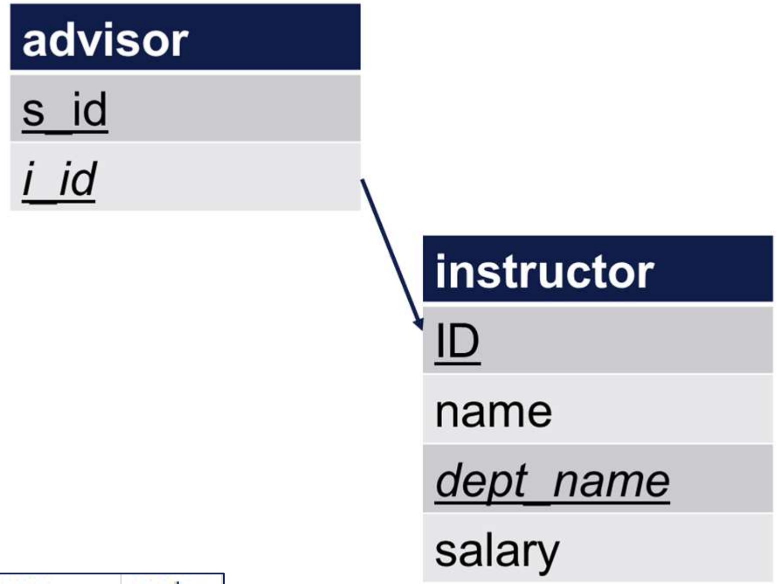
Let us consider the previous query:

- For all the instructors in the university that are an advisor for some students, find their names and the student ids that they are an advisor for



select *name* ***as*** *instructor_name, s_id*
from *instructor, advisor*
where *instructor.ID=advisor.i_id;*

	instructor_name	s_id
1	Sara	1000
2	Leonardo	1002

select *T.name, A.s_id*
from *instructor* ***as*** *T, advisor* ***as*** *A*
where *T.ID=A.i_id;*

	name	s_id
1	Sara	1000
2	Leonardo	1002

Additional Operations: order by

Let us consider the following query:

- Retrieve the name of all instructors in the Computing department and sort their names alphabetically

select *name*
from *instructor*
where *dept_name='Computing'*
order by *name;*

	name
1	Leonardo
2	Sara

instructor
<u>ID</u>
name
<u>dept_name</u>
salary

Additional Operations: order by

Let us consider the following query:

- Retrieve the name of all instructors in the Computing department and sort their names alphabetically

select *name*
from *instructor*
where *dept_name='Computing'*
order by *name;*

- Include also their salary in descending order

instructor
<u>ID</u>
name
<u>dept_name</u>
salary

Additional Operations: order by

Let us consider the following query:

- Retrieve the name of all instructors in the Computing department and sort their names alphabetically

> ***select*** *name*
> ***from*** *instructor*
> ***where*** *dept_name='Computing'*
> ***order by*** *name;*

instructor
<u>ID</u>
name
<u>*dept_name*</u>
salary

- Include also their salary in descending order

> ***select*** *name, salary*
> ***from*** *instructor*
> ***where*** *dept_name='Computing'*
> ***order by*** *salary* ***desc****, name* ***asc****;*

	name	salary
1	Sara	64000
2	Leonardo	55000

Additional Operations: limit

Let us consider the following query:

- Retrieve the name and id of the top 10 instructors in the Computing department for highest salary

instructor
ID
name
dept_name
salary

Additional Operations: limit

Let us consider the following query:

- Retrieve the name and id of the top 10 instructors in the Computing department for highest salary

> ***select*** *name, ID*
> ***from*** *instructor*
> ***where*** *dept_name='Computing'*
> ***order by*** *salary* ***desc***
> ***limit*** *10;*

instructor
<u>ID</u>
name
<u>dept_name</u>
salary

Overview

- Recap
- Advanced Queries
- **Data Definition**
- Schema Definition

Basic Types

The SQL standard supports a variety of built-in types, including:

- ***char***(*n*): A fixed-length character of user-specified length *n*
- ***varchar***(*n*): A variable-length character of user-specified maximum length *n*

Basic Types

The SQL standard supports a variety of built-in types, including:

- ***int****:* An integer (machine dependent). The full form ***integer*** is equivalent
- ***smallint****:* A small integer (machine dependent)

Basic Types

The SQL standard supports a variety of built-in types, including:

- ***numeric****(p,d):* A fixed-point number with user-specified precision. The number consists of *p* digits (plus the sign), and *d* of the *p* digits are to the right of the decimal point

- ***real, double precision****:* (double) floating-point number (machine dependent)

- ***float****(n):* floating-point number with precision of at least *n* digits

Overview

- Recap
- Advanced Queries
- Data Definition
- **Schema Definition**

Schema Definition

We define a new SQL relation by using the **create table** command:

create table *r*

$(A_1 \quad D_1,$

$A_2 \quad D_2,$

$\ldots,$

$A_n \quad D_n,$

<integrity_constraint$_1$*>,*

…,

<integrity_constraint$_2$*>);*

Schema Definition

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- *r* – name of the relation

Schema Definition

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<integrity_constraint$_2$*>);*

- *r* – name of the relation
- A_i and D_i – A_i is the name of an attribute and D_i is the domain of attribute A_i

Schema Definition

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- *r* – name of the relation
- A_i and D_i – A_i is the name of an attribute and D_i is the domain of attribute A_i
- *integrity_constraint* - **primary key, foreign key, not null**

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Schema Definition

Let's look at the definition of the *department* table:

```
create table department
        (dept_name varchar (20),
        building   varchar (15),
        budget     numeric (12,2),
        primary key (dept_name));
```

department
dept_name
building
budget

Schema Definition

What about the *instructor* table?

instructor
ID
name
dept_name
salary

department
dept_name
building
budget

Schema Definition

What about the *instructor* table?

> ***create table*** *instructor*
> *(ID* ***int****,*
> *name* ***varchar*** *(20)* ***not null****,*
> *dept_name* ***varchar*** *(20),*
> *salary* ***numeric*** *(8,2),*
> ***primary key*** *(ID),*
> ***foreign key*** *(dept_name)* ***references*** *department);*

instructor
ID
name
dept_name
salary

department
dept_name
building
budget

Aims of the Session

You should now be able to:

1. Understand the basic principles of SQL
2. Describe the different data types in SQL
3. Apply the theoretical concepts to complete SQL queries and schema definition queries

References

Silberschatz, A., Korth, H. F., and Sudarshan, S. (2010), *Database System Concepts*, 6th Edition. McGraw-Hill.

- Chapter 3 – Introduction to SQL (up to Section 3.4.5)